

HEMANTH IRIVICHETTY

Machine Learning & AI Engineer

+91 8500363606 | ihemanth.2001@gmail.com | linkedin.com/in/hemanth-irivichetty | github.com/Hemanth-stack

PROFESSIONAL SUMMARY

AI and MLOps Engineer with 3+ years of experience shipping production-grade NLP and LLM systems at scale. At Zoho Corporation, I owned both sides of the AI stack building in-house NLP features (grammar correction, summarization, question answering, and NL-to-SQL analytics) and engineering the inference platform that powered them delivering a 4x throughput increase, 40% latency reduction, and 60% cost savings. I implemented tensor parallelism for multi-GPU inference. At Arrise Solutions, I build computer vision and audio intelligence systems for live game-stream support designing pose-estimation pipelines for real-time posture tracking of on-stream personnel and developing audio-monitoring models for continuous sound-event detection and speech analysis across simultaneous live iGaming feeds.

TECHNICAL SKILLS

Core Languages : Python (AsyncIO), Bash, SQL

ML Frameworks : PyTorch, Hugging Face Transformers, PEFT, scikit-learn, NumPy, Pandas

NLP Tasks : Grammar Correction, Text Summarization, Question Answering, NL-to-SQL / NLA, Sequence Labeling

LLM & Fine-Tuning : LLaMA 3.1, Qwen 2.5, FLAN-T5, LoRA, QLoRA, Prompt Engineering, RLHF (exposure)

LLM Inference : vLLM, PagedAttention, Continuous Batching, CTranslate2, Tensor Parallelism

RAG & Agents : LangChain, FAISS, Hybrid Search, Cross-Encoder Re-ranking, CrewAI, Multi-Agent Systems

MLOps & CI/CD : Kubernetes (EKS), Docker, MLflow, Prometheus, Grafana, DVC

Cloud & Infra : Azure (OpenAI), GCP (Vertex AI), Distributed Systems, Microservices

Backend & Security : FastAPI, Celery, SQLAlchemy (Async), REST APIs, JWT / OAuth2, Hybrid Encryption (RSA + Fernet)

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Apr 2026 – Present

Arrise Solutions (India) Pvt Ltd · India · Remote

- ▶ **Domain:** Building and deploying ML/CV models for gaming and iGaming analytics, automating data-intensive processes that were previously performed manually.
- ▶ **Computer Vision:** Designing and experimenting with posture-tracking and image-analysis pipelines using PyTorch, OpenCV, and YOLO to extract actionable intelligence from gaming streams and visual data.
- ▶ **MLOps Infrastructure:** Establishing model lifecycle management with MLflow and DVC; containerizing workloads with Docker and Kubernetes for reproducible, scalable experimentation and deployment workflows.

LLM & Vision Infrastructure Engineer

Jul 2025 – Mar 2026

Logos Technologies LLC (Eonforge) · Dubai, UAE · Remote

- ▶ **System Architecture:** Designed AI infrastructure for LumenCipher, an enterprise insurance CRM spanning RAG pipelines, OCR/vision systems, and multi-agent orchestration, all secured with JWT/OAuth2 authentication.
- ▶ **Hybrid RAG Pipeline:** Built a production RAG system combining FAISS vector search with knowledge-graph traversal for multi-hop reasoning; added Cross-Encoder re-ranking (BGE) to boost retrieval precision, with end-to-end evaluation via BLEU and ROUGE metrics.
- ▶ **Document Intelligence:** Engineered an OCR and vision pipeline using Tesseract with custom layout analysis and YOLO-based object detection, enabling automated structured-data extraction from complex insurance claim forms at hundreds of documents per day.

- ▶ **Agent Orchestration:** Architected the multi-agent workflow layer that automates claims processing end-to-end, integrating AI agents with the CRM database via SQLAlchemy Async and significantly reducing manual intervention in the claims lifecycle.

Member of Technical Staff NLP & AI

Jun 2023 – Jun 2025

Zoho Corporation · Chennai, India · On-site

- ▶ **NLP Feature Ownership:** Built and shipped four customer-facing NLP capabilities on a unified AI platform: grammar correction, abstractive summarization, extractive question answering, and Natural Language Analytics (NL-to-SQL) using fine-tuned FLAN-T5, LLaMA 3.1, and custom in-house models via LoRA/QLoRA. Evaluated with BLEU/ROUGE scores and human review cycles.
- ▶ **NL-to-SQL / NLA Workflows:** Developed natural-language query interfaces over Zoho's internal databases and BI dashboards, enabling business users to query structured data in plain English; built schema-aware SQL generation with intent parsing, achieving strong query success rates on internal benchmarks.
- ▶ **Model Fine-Tuning:** Fine-tuned FLAN-T5 and early LLaMA variants on domain-specific datasets using LoRA/QLoRA and PEFT, developing foundational expertise in low-resource adaptation.
- ▶ **4x Throughput Optimization:** Migrated the shared inference platform powering all 4 NLP features from a custom FastAPI queue to vLLM with Continuous Batching and PagedAttention, scaling sustained throughput 4x under concurrent multi-feature load.
- ▶ **Distributed Inference:** Implemented tensor parallelism and model sharding for multi-GPU inference, allowing larger models to be served reliably under sustained high-concurrency production traffic.
- ▶ **Security:** Secured on-premise-to-cloud model communication using hybrid encryption (RSA + Fernet symmetric keys), ensuring data integrity across the inference boundary.

Project Trainee AI/ML

Oct 2022 – May 2023

Zoho Corporation · Chennai, India · On-site

- ▶ **Architecture Migration:** Led the migration of core NLP workflows from RNN/LSTM architectures to Transformer-based pipelines, significantly improving sequence-labeling accuracy across production datasets.
- ▶ **Data Engineering:** Built annotation interfaces and data-migration pipelines using Zoho Catalyst, streamlining preprocessing of large-scale text corpora for model-training jobs.

NOTABLE PROJECTS

GEC Grammar Error Correction System

End-to-end grammar-correction pipeline fine-tuned on FLAN-T5 using LoRA over domain-specific text corpora. Includes a custom preprocessing pipeline, a BLEU-based evaluation harness, and a FastAPI inference server demonstrating low-resource fine-tuning and production-ready NLP serving.

OLLM Optimized LLM Inference Server

A lightweight, production-oriented LLM inference server built on top of vLLM with custom batching strategies, Int8 quantization support, and a Prometheus metrics endpoint. Designed to benchmark throughput/latency trade-offs across quantization and batching configurations.

EDUCATION

B.Tech, Computer Science & Engineering

Aug 2019 – Jul 2023

Jawaharlal Nehru Technological University Anantapur (JNTUA) · CGPA: 7.72